A MACHINE LEARNING-BASED VIRTUAL SCREENING TOOL TO IDENTIFY SELECTIVE MT1-MMP INHIBITORS FOR THE TREATMENT OF OSTEOSARCOMA

Osteosarcoma (OS) is a type of malignant cancer related to the irregular growth and proliferation of bone-forming cells. While there are a few available chemotherapeutic methods for its treatment (e.g., US FDA-approved LEVO leucovorin injection along with methotrexate), most are associated with toxic side effects and require high doses. Membrane-type-1 Matrix Metalloproteinase (MT1-MMP) is found to be overexpressed on the affected cell surface matrix along with other MMPs. Thus, the inhibitors of MT1-MMP or MMP-14 can play a significant role in the treatment of OS by reducing the invasion and metastasis of the tumor cells. But the available MMPIs suffer from a lack of specificity and are associated with off-target toxicity due to high homology among the structures of MMPs. A machine learning model was developed through ATOM (Accelerating Therapeutics for Opportunities in Medicine) is a platform for accelerating the drug discovery process using computational modelling to effectively screen for potent and specific MMP-14 inhibitors from the available literature data. These models gave more accurate and precise information about the selectivity of MMP-14 inhibitors over the inhibitors of other metalloproteinases, with a substantial increase in inhibition performance.